

D. Division Versus Addition

time limit per test: 2 seconds

memory limit per test: 1024 megabytes

For an array $b = [b_1, b_2, \dots, b_m]$ of length m ($b_i \geq 2$), consider the following two-player game played by Poby and Rekkles.

- The players take turns, with Poby moving first.
- On Poby's turn, he must choose an element $x \geq 2$ and replace it with $\left\lfloor \frac{x}{2} \right\rfloor$. In other words, he picks i ($1 \leq i \leq m$) such that $b_i \geq 2$, then does $b_i := \left\lfloor \frac{b_i}{2} \right\rfloor$.
- On Rekkles' turn, he must choose an element $x \geq 2$ from the array b and replace it with $x + 1$. In other words, he picks i ($1 \leq i \leq m$) such that $b_i \geq 2$, then does $b_i := b_i + 1$.

The game ends once all elements in the array b are equal to 1.

Define the **score** of the game as the number of moves that Poby makes. Poby's goal is to minimize the **score**, while Rekkles's goal is to maximize the **score**.

Then, the **value** of the array b is the score of the game when both players play optimally.

You are given an integer array a of length n ($a_i \geq 2$).

Answer q independent queries. In each query, you are given a range $1 \leq l \leq r \leq n$ and must find the **value** of the array $[a_l, a_{l+1}, \dots, a_r]$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \leq n, q \leq 250\,000$) — the length of the array and the number of queries.

The next line contains n integers $a_1, a_2, \dots, a_n$ ($2 \leq a_i \leq 10^9$) — the elements of the array a .

Then q lines follow. The j -th of them contains two integers l_j and r_j ($1 \leq l_j \leq r_j \leq n$) — the range of the subarray for the i -th query.

It is guaranteed that the sum of n over all test cases does not exceed 250 000.

It is guaranteed that the sum of q over all test cases does not exceed 250 000.

Output

For each test case, output q lines. The i -th line should contain a single integer representing the answer to the i -th query.

Example

input

Copy

```
2
5 5
4 3 2 5 6
1 1
1 2
2 4
3 5
1 5
10 1
314 159 265 358 979 323 846 264 338 327
1 10
```

Squarepoint Challenge (Codeforces Round 1055, Div. 1 + Div. 2)

Contest is running

01:00:30

Contestant



→ Submit?

Language: Python 3.13.2 
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	342
Problem B	1024
Problem C	1024
Problem D	1707
Problem E	2048
Problem F	2731
Problem G	2731
Problem H1	2731
Problem H2	1707
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 01:59 from the first attempt

↑ output

[Copy](#)

```
2
3
5
6
10
91
```

Note

Explanation of the first test case, first query (1 1):

The subarray is [4].

1. Poby: $4 \rightarrow \left\lfloor \frac{4}{2} \right\rfloor = 2$. The array is [2].
2. Rekkles: $2 \rightarrow 3$. The array is [3].
3. Poby: $3 \rightarrow \left\lfloor \frac{3}{2} \right\rfloor = 1$. The array is [1], so the game ends.

It can be shown that this strategy is optimal for both players. Therefore, the value of the array [4] is 2.

Explanation of the first test case, second query (1 2):

The subarray is [4, 3].

1. Poby: $3 \rightarrow \left\lfloor \frac{3}{2} \right\rfloor = 1$. The array is [4, 1].
2. Rekkles: $4 \rightarrow 5$. The array is [5, 1].
3. Poby: $5 \rightarrow \left\lfloor \frac{5}{2} \right\rfloor = 2$. The array is [2, 1].
4. Rekkles: $2 \rightarrow 3$. The array is [3, 1].
5. Poby: $3 \rightarrow \left\lfloor \frac{3}{2} \right\rfloor = 1$. The array is [1, 1], so the game ends.

It can be shown that this strategy is optimal for both players. Therefore, the value of the array [4, 3] is 3.

